Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- jqassistant
- Neo4j Python Driver

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Change]	85144	78.288294
1	[Git, Commit]	11055	10.164863
2	[File, Git]	5704	5.244720
3	[Git, Tag]	1644	1.511627
4	[Author, Git, Person]	1253	1.152110
5	[Json, Key]	668	0.614213
6	[Json, Value, Scalar]	603	0.554447
7	[Committer, Git, Person]	370	0.340208
8	[NPM, Dependency]	338	0.310785
9	[Type, TS, Primitive]	291	0.267569
10	[Type, TS, Declared]	277	0.254696
11	[TS, ExternalDeclaration]	215	0.197688
12	[Type, TS, Literal]	136	0.125049
13	[Json, Value, Object]	133	0.122291
14	[Type, TS, Union]	119	0.109418
15	[Type, TS, ObjectMember]	102	0.093787
16	[NPM, Script]	91	0.083673
17	[TS, Property]	65	0.059766
18	[TS, Function]	47	0.043216
19	[Type, TS, FunctionParameter]	40	0.036779
20	[Type, Object, TS]	39	0.035860
21	[Git, Branch]	39	0.035860
22	[File, Directory]	34	0.031262
23	[Type, TS, Function]	34	0.031262
24	[TS, Parameter]	33	0.030343
25	[Package, File, Json, NPM]	29	0.026665
26	[TS, Variable]	24	0.022068
27	[Value, TS, Literal]	20	0.018390
28	[jQAssistant, Rule, Concept]	19	0.017470
29	[Type, TS, Intersection]	17	0.015631

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

```
<Figure size 640x480 with 0 Axes>
```

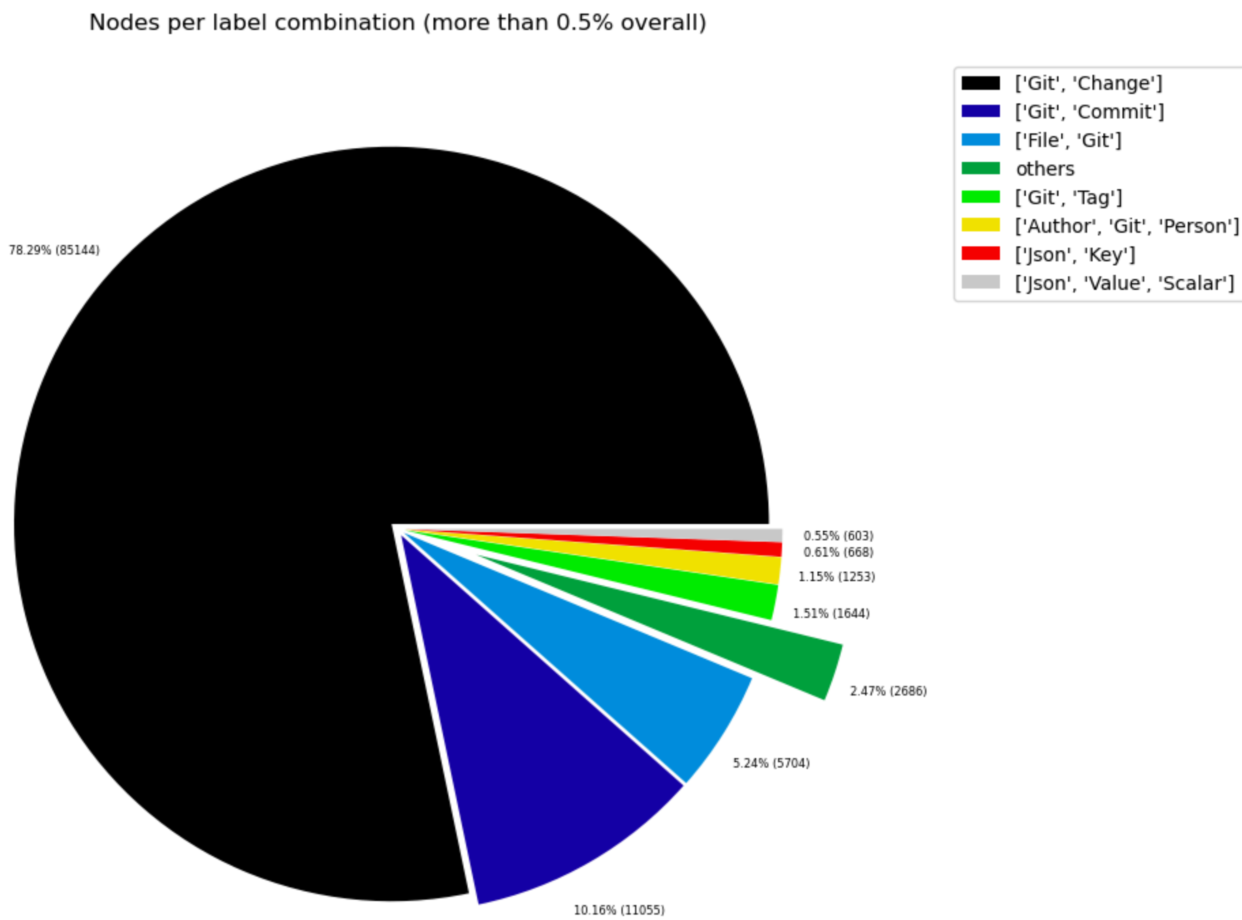

Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000919
1	[File, TS, Scan]	1	0.000919
2	[TS, Method]	1	0.000919
3	[Repository, File, Git]	1	0.000919
4	[TS, Constructor]	1	0.000919
5	[Value, TS, ObjectMember]	1	0.000919
6	[TS, Class]	1	0.000919
7	[TS, Enum]	2	0.001839
8	[Value, Object, TS]	3	0.002758
9	[Type, TS, Tuple]	3	0.002758
10	[Value, TS, Function]	4	0.003678
11	[TS, TypeParameter]	4	0.003678
12	[Value, TS, Complex]	5	0.004597
13	[NPM, Engine]	6	0.005517
14	[Project, TS]	6	0.005517
15	[File, Local]	6	0.005517
16	[Value, TS, Call]	6	0.005517
17	[Value, TS, Member]	6	0.005517
18	[File, TS, Local, Module]	6	0.005517
19	[Type, TS, TypeParameterReference]	6	0.005517
20	[TS, EnumMember]	8	0.007356
21	[Type, TS, NotIdentified]	11	0.010114
22	[TS, ExternalModule]	11	0.010114
23	[Json, Value, Array]	12	0.011034
24	[Value, TS, Declared]	13	0.011953
25	[TS, TypeAlias]	16	0.014712
26	[File, Directory, Local]	16	0.014712
27	[Type, TS, Intersection]	17	0.015631
28	[TS, Interface]	17	0.015631
29	[jQAssistant, Rule, Concept]	19	0.017470

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

```
<Figure size 640x480 with 0 Axes>
```

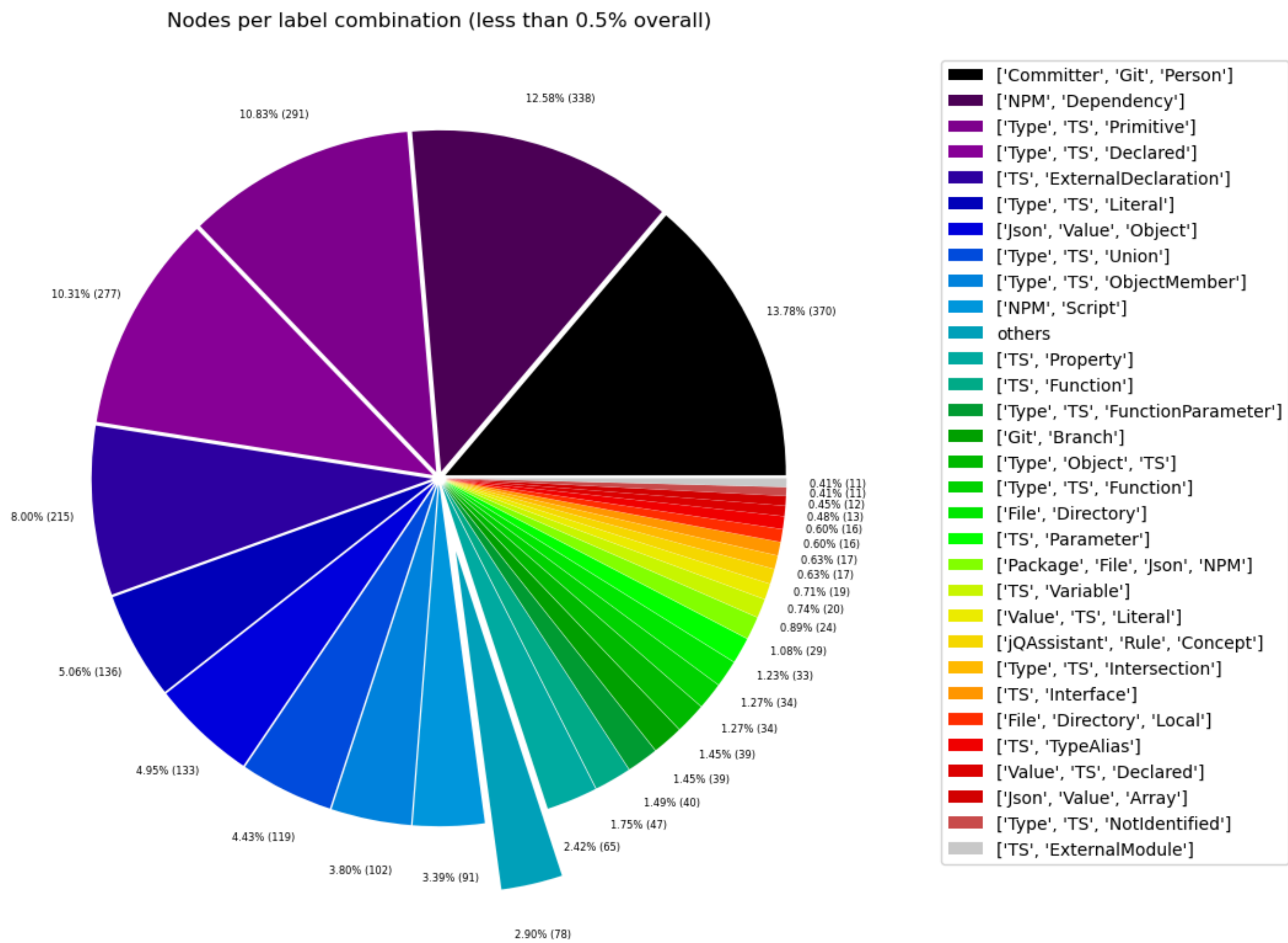

Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	105210	96.738601
1	Change	85144	78.288294
2	Commit	11055	10.164863
3	File	5797	5.330232
4	Tag	1644	1.511627
5	Person	1623	1.492318
6	TS	1591	1.462894
7	Json	1445	1.328650
8	Author	1253	1.152110
9	Type	1075	0.988442
10	Value	806	0.741102
11	Key	668	0.614213
12	Scalar	603	0.554447
13	NPM	464	0.426639
14	Committer	370	0.340208
15	Dependency	338	0.310785
16	Primitive	291	0.267569
17	Declared	290	0.266650
18	ExternalDeclaration	215	0.197688
19	Object	175	0.160909
20	Literal	156	0.143439
21	Union	119	0.109418
22	ObjectMember	103	0.094707
23	Script	91	0.083673
24	Function	85	0.078156
25	Property	65	0.059766
26	Directory	50	0.045974
27	FunctionParameter	40	0.036779
28	Branch	39	0.035860
29	Parameter	33	0.030343
30	Package	29	0.026665
31	Local	28	0.025745
32	Variable	24	0.022068
33	jQAssistant	20	0.018390
34	Concept	19	0.017470
35	Rule	19	0.017470
36	Interface	17	0.015631
37	Intersection	17	0.015631
38	TypeAlias	16	0.014712
39	Array	12	0.011034

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

```
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```
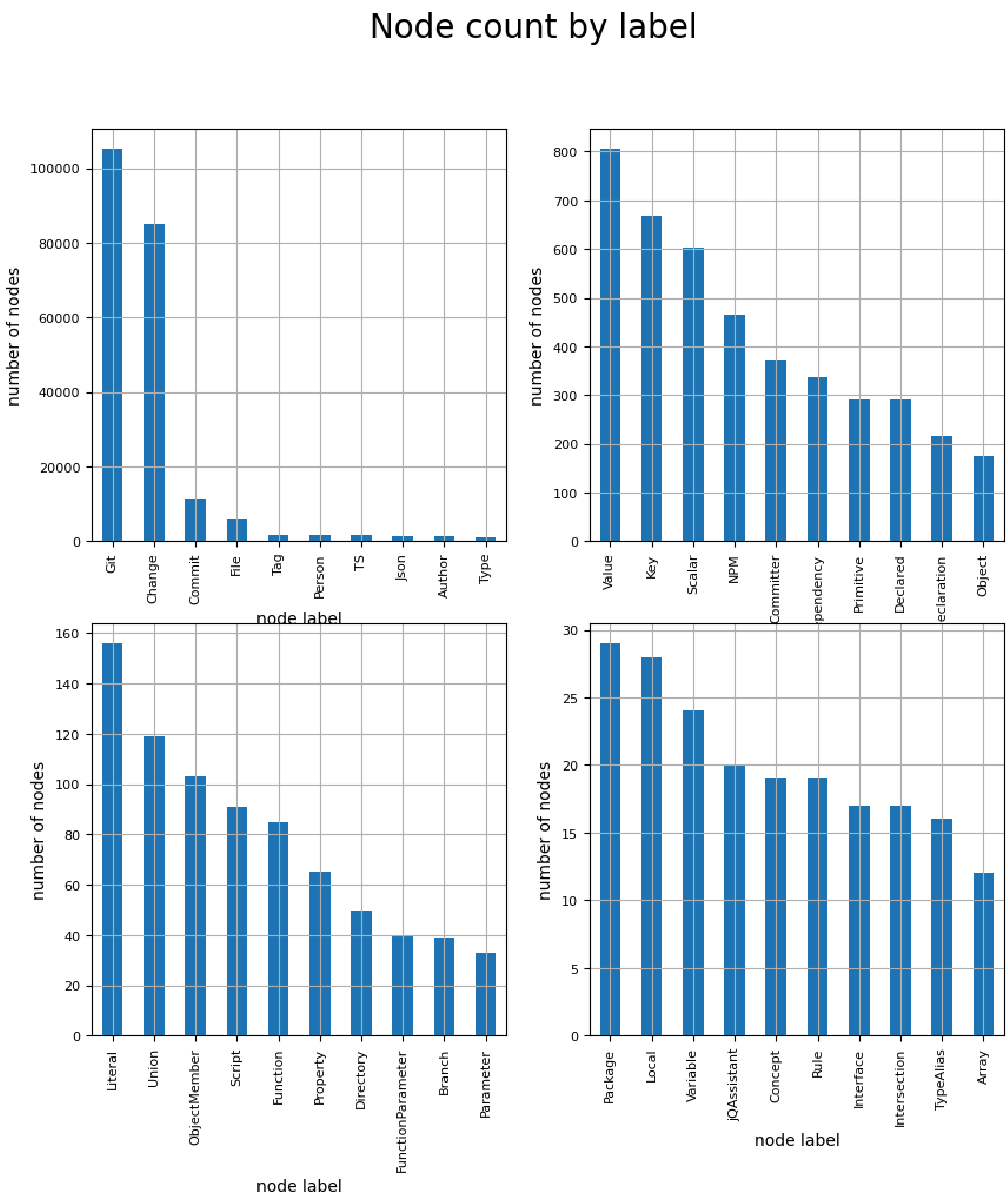



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

```
Total number of relationships: 324772
```

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	85144	26.216546
1	MODIFIES	85144	26.216546
2	UPDATES	56632	17.437464
3	COMMITTED	22110	6.807853
4	CREATES	19881	6.121525
5	HAS_PARENT	12122	3.732465
6	DELETES	11910	3.667188
7	HAS_COMMIT	11055	3.403926
8	HAS_FILE	5704	1.756309
9	RENAMES	3279	1.009631
10	HAS_NEW_NAME	1763	0.542842
11	HAS_TAG	1644	0.506201
12	ON_COMMIT	1644	0.506201
13	HAS_AUTHOR	1253	0.385809
14	DEPENDS_ON	887	0.273115
15	HAS_KEY	668	0.205683
16	HAS_VALUE	668	0.205683
17	CONTAINS	594	0.182898
18	HAS_COMMITTER	370	0.113926
19	OF_TYPE	338	0.104073
20	EXPORTS	309	0.095144
21	REFERENCES	197	0.060658
22	DECLARES	186	0.057271
23	DECLARES_DEV_DEPENDENCY	169	0.052037
24	DECLARES_DEPENDENCY	161	0.049573
25	HAS_MEMBER	103	0.031715
26	HAS_TYPE_ARGUMENT	94	0.028943
27	DECLARES_SCRIPT	91	0.028020
28	RETURNS	82	0.025248
29	COPIES	81	0.024941

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

```
<Figure size 640x480 with 0 Axes>
```

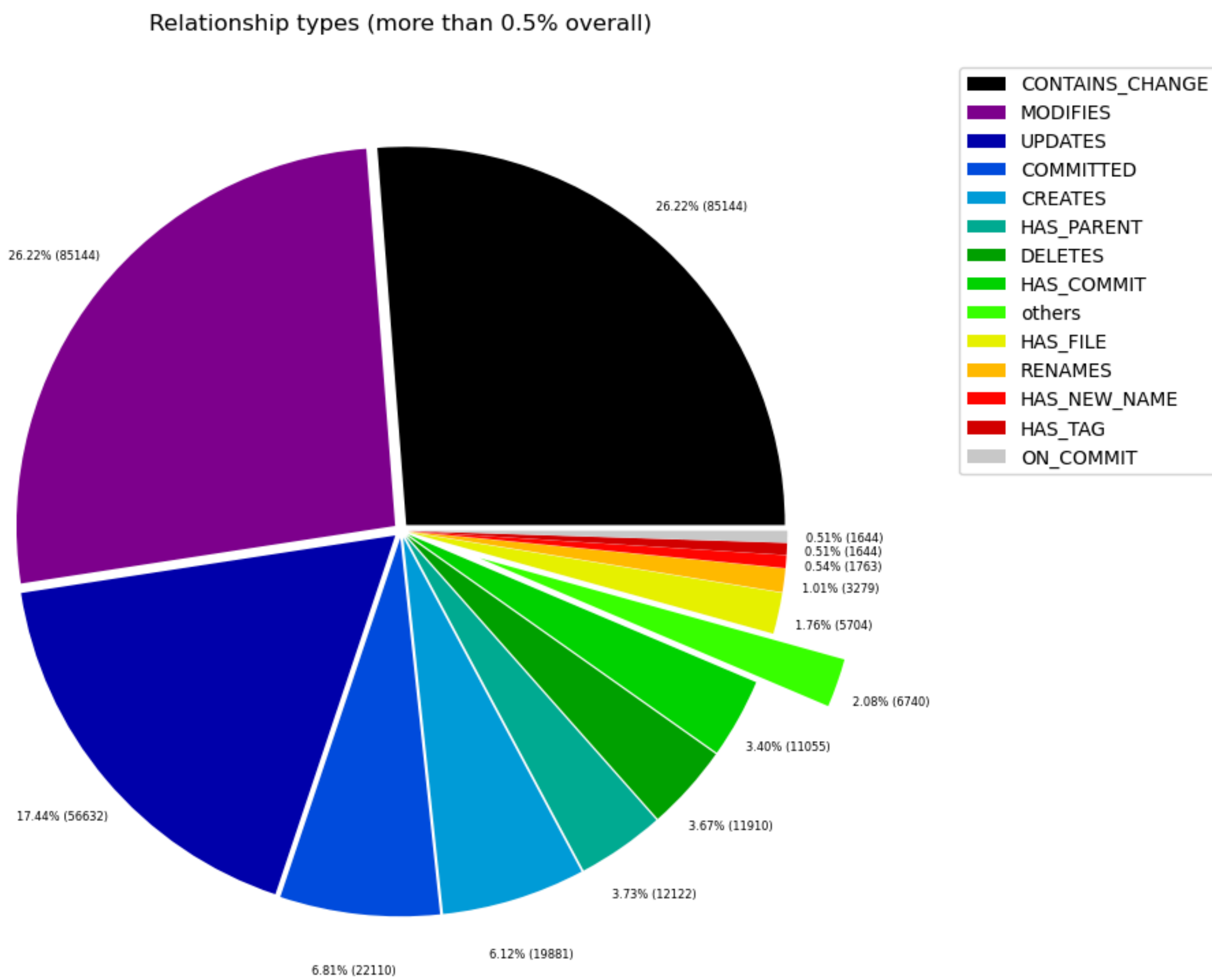


Table 2b - Lowest relationship count by type

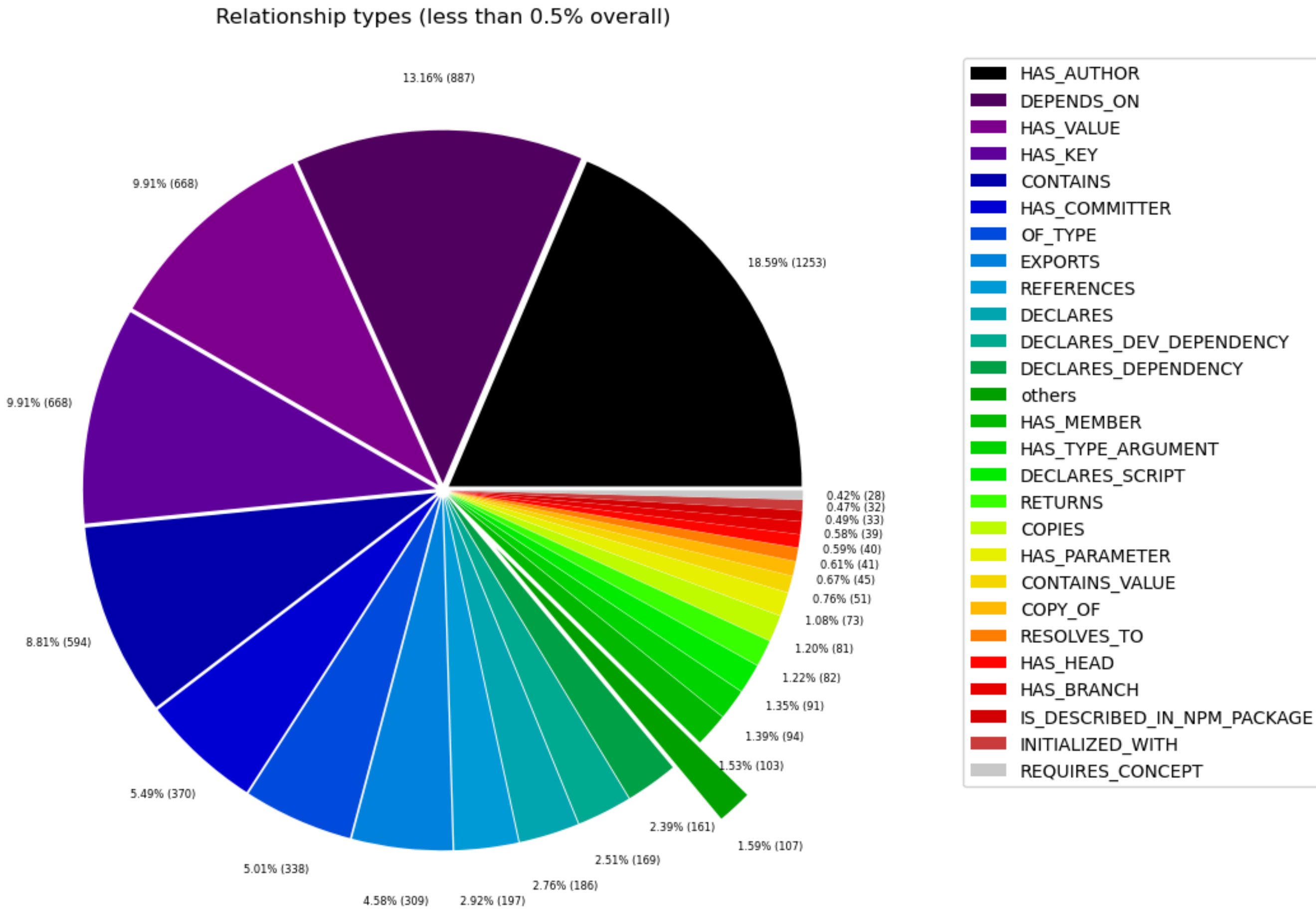
Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONSTRAINED_BY	4	0.001232
1	REFERENCED_PROJECTS	5	0.001540
2	MEMBER	6	0.001847
3	HAS_ROOT	6	0.001847
4	HAS_NPM_PACKAGE	6	0.001847
5	HAS_CONFIG	6	0.001847
6	HAS_ARGUMENT	6	0.001847
7	EXTENDS	6	0.001847
8	DECLARES_ENGINE	6	0.001847
9	CONTAINS_PROJECT	6	0.001847
10	CALLS	6	0.001847
11	PARENT	6	0.001847
12	DECLARES_PEER_DEPENDENCY	8	0.002463
13	USES	11	0.003387
14	INCLUDES_CONCEPT	19	0.005850
15	REQUIRES_CONCEPT	28	0.008621
16	INITIALIZED_WITH	32	0.009853
17	IS_DESCRIBED_IN_NPM_PACKAGE	33	0.010161
18	HAS_BRANCH	39	0.012008
19	HAS_HEAD	40	0.012316
20	RESOLVES_TO	41	0.012624
21	COPY_OF	45	0.013856
22	CONTAINS_VALUE	51	0.015703
23	HAS_PARAMETER	73	0.022477
24	COPIES	81	0.024941
25	RETURNS	82	0.025248
26	DECLARES_SCRIPT	91	0.028020
27	HAS_TYPE_ARGUMENT	94	0.028943
28	HAS_MEMBER	103	0.031715
29	DECLARES_DEPENDENCY	161	0.049573

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

```
<Figure size 640x480 with 0 Axes>
```



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberOf
0	[Git, Change]	MODIFIES	[File, Git]	85144	85144	
1	[Git, Commit]	CONTAINS_CHANGE	[Git, Change]	85144	11055	
2	[Git, Change]	UPDATES	[File, Git]	56632	85144	
3	[Git, Change]	CREATES	[File, Git]	19881	85144	
4	[Git, Commit]	HAS_PARENT	[Git, Commit]	12122	11055	
5	[Git, Change]	DELETES	[File, Git]	11910	85144	
6	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11055	1	
7	[Author, Git, Person]	COMMITTED	[Git, Commit]	11055	1253	
8	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11055	370	
9	[Repository, File, Git]	HAS_FILE	[File, Git]	5704	1	
10	[Git, Change]	RENAMES	[File, Git]	3279	85144	
11	[File, Git]	HAS_NEW_NAME	[File, Git]	1763	5704	
12	[Repository, File, Git]	HAS_TAG	[Git, Tag]	1644	1	
13	[Git, Tag]	ON_COMMIT	[Git, Commit]	1644	1644	
14	[Repository, File, Git]	HAS_AUTHOR	[Author, Git, Person]	1253	1	
15	[Json, Value, Object]	HAS_KEY	[Json, Key]	668	133	
16	[Json, Key]	HAS_VALUE	[Json, Value, Scalar]	552	668	
17	[Repository, File, Git]	HAS_COMMITTER	[Committer, Git, Person]	370	1	
18	[TS, Function]	DEPENDS_ON	[TS, ExternalDeclaration]	293	47	
19	[File, TS, Local, Module]	DEPENDS_ON	[TS, ExternalDeclaration]	236	6	
20	[TS, ExternalModule]	EXPORTS	[TS, ExternalDeclaration]	215	11	
21	[Package, File, Json, NPM]	DECLARES_DEV_DEPENDENCY	[NPM, Dependency]	169	29	
22	[Package, File, Json, NPM]	DECLARES_DEPENDENCY	[NPM, Dependency]	161	29	
23	[Type, TS, Union]	CONTAINS	[Type, TS, Primitive]	147	119	
24	[Type, TS, Declared]	REFERENCES	[TS, ExternalDeclaration]	143	277	
25	[Type, TS, Union]	CONTAINS	[Type, TS, Literal]	119	119	
26	[Json, Key]	HAS_VALUE	[Json, Value, Object]	104	668	
27	[Type, Object, TS]	HAS_MEMBER	[Type, TS, ObjectMember]	102	39	
28	[Package, File, Json, NPM]	DECLARES_SCRIPT	[NPM, Script]	91	29	
29	[Git, Change]	COPIES	[File, Git]	81	85144	

Graph Density

```
total_number_of_nodes (vertices): 108757
total_number_of_relationships (edges): 324772
-> total directed graph density: 2.7457951553357747e-05
-> total directed graph density in percent: 0.0027457951553357747
```